



OGHENEFEGOR FAVOUR UGBINE

Graduate Mechanical Engineer · FEA & Design Validation · Maintenance & Reliability
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Immediately Available · Open to Work in France · Category B Driving Licence

PROFILE

Versatile mechanical engineer with an Erasmus Mundus Joint MSc in Manufacturing 4.0 and a Bachelor of Engineering in Mechanical Engineering, bringing 3+ years across mechanical design, FEM simulation, and maintenance & reliability, underpinned by hands-on health, safety and environment (HSE) and fire-safety experience in classified explosion-risk facilities. Skilled in SolidWorks 3D CAD and GD&T, static and dynamic FEA in Abaqus, and machining simulation including a Python-Abaqus FEM pipeline that cut simulation-to-experiment error to below 0.5%. On the maintenance side, experienced in diagnostic, preventive, and corrective maintenance of safety-critical mechanical systems, FMEA-driven root-cause analysis, and thorough technical documentation and traceability [using CMMS-based maintenance workflows]. Continuously advancing in manufacturing as a Lean Six Sigma Black Belt, with a commitment to embedding sustainable industrial practice from biodegradable material selection to life-cycle-conscious design. Named a top 5 global finalist (4th place, \$50,000 team award) in the Sanofi 2025 Device Innovation Challenge for CAD-led design of a wearable drug-delivery enclosure and fluid path, with attention to plastic-component manufacturability and ergonomics. Structured, safety-first, and effective in multidisciplinary, multinational teams across design, manufacturing, quality, and maintenance.

CORE COMPETENCIES

SolidWorks 3D Modelling & GD&T	Equipment troubleshooting	Mechanical Calculations & Sizing	FEM / Structural Analysis
Preventive & Corrective Maintenance	Manufacturing Methods & DFM/DFMEA	Technical Documentation, Specs & BOMs	Multinational Team Collaboration
CMMS & Asset management	Explosion-risk / HSE Safety	Continuous Improvement (FMEA/Kaizen)	EMEA Mobility · English C1 · French B1

EDUCATION

Erasmus Mundus Joint MSc in Manufacturing 4.0 - META 4.0

Sept. 2023 – Sept. 2025

Ecole Centrale de Lyon · Ecole des Mines de Saint-Étienne · Politecnico di Torino · France & Italy

- **European Commission full scholarship** - selected among the top global 1%; one of only 25 scholars in the cohort
- **MSc Mechanical Engineering (Ecole Centrale de Lyon)**: Specialising in Sustainable Manufacturing, **MSc Materials Engineering for Industry 4.0 (Politecnico di Torino)**, and **MSc Materials Engineering for Industry 4.0 (Ecole des Mines de Saint-Étienne)**
- **Relevant coursework**: mechanical component design, continuum mechanics, constitutive modelling, additive manufacturing & 3D printing, materials forming, machining simulation, surface integrity, life cycle assessment, functional and in-use properties of materials
- **Master's Thesis**: Designed and executed a Python–Abaqus FEM computational pipeline to inversely identify Johnson–Cook material constitutive parameters for machining simulation; reduced prediction errors from 40% to under 1% within 5 iterations

Bachelor of Engineering in Mechanical Engineering

Aug. 2016 – Jan. 2022

Cross River University of Technology (CRUTECH) · Nigeria

- **CGPA 3.77/5.00** Second Class Honours Upper Division · Ranked 4th of 124 graduates (Top 3.2%)
- **Relevant coursework**: engineering mechanics, machine design, thermodynamics, fluid mechanics, manufacturing technology, automotive technology, strength of materials, applied mathematics.

PROFESSIONAL EXPERIENCE

Machining Simulation Researcher (Master's Thesis Intern)

March 2025 – Sept. 2025

MISUTECH · Saint-Etienne, France

- Designed detailed 3D FEM models of orthogonal cutting operations in Abaqus; defined component geometry, contact interfaces, mesh configurations, and thermal boundary conditions for a stainless-steel machining process
- Conducted technical analyses and convergence calculations to quantify simulation accuracy against experimental observables - chip thickness, cutting force, contact length, and penetration force across five chip thickness configurations
- Developed a Python–Abaqus automation pipeline coupling FEM simulation outputs to a Levenberg–Marquardt optimisation algorithm, eliminating manual iteration and reducing average prediction errors from 40% to under 0.5% within 5 cycles
- Extended and validated calibrated parameter sets from 2D orthogonal cutting to full 3D FEM configurations on a second stainless steel alloy, confirming model generalizability for industrial virtual machining
- Produced full engineering documentation covering model assumptions, mesh sensitivity studies, iterative parameter revisions, validation outcomes, and handover materials, enabling traceability and reproducibility

Maintenance Engineer

Aug. 2022 – Aug. 2023

Techno Oil Limited · Lagos, Nigeria

- Maintained, inspected, and overhauled automated blending, filling, and packaging line machinery in a high-throughput LPG energy production facility; conducted component tolerance checks, seal inspections, and functional tests on mechanical assemblies
- Managed and validated safety-critical mechanical systems: LPG process lines, vented air compressors, fire pump engines, and water treatment equipment, including pressure testing, seal integrity verification, and valve performance testing against operational standards
- Applied systematic root-cause analysis and FMEA thinking to diagnose recurring faults, document corrective actions, and implement preventive measures that sustain continuous line reliability
- Collaborated with electronics engineers, process operators, and HSE teams to prioritize corrective actions and restore performance under tight operational timelines

Automotive Engineer (Diagnostics and Maintenance)

March 2021 – May 2022

- Performed component-level technical analysis of mechanical and electrical faults across complex automotive systems; used diagnostic tools, tolerance measurement, and structured root-cause methodology to identify and resolve failures
- Developed and maintained preventive maintenance schedules that reduced unplanned downtime; integrated quality control checkpoints to improve first-time fix rates and system longevity

Fire and Safety Engineer Intern

Oct. 2019 – March 2020

Nigerian Gas Company Limited (NNPC Subsidiary) · Warri, Nigeria

- Deployed to the Asset Management Delta Department of a national petroleum company operating classified explosion-risk and hazardous-area facilities
- Performed pressure testing and operational validation of fire hydrant systems, standpipes, and fire pump engines in accordance with safety regulations and industry standards; conducted daily HSE compliance audits across process areas
- Delivered fire and explosion-risk safety training to plant personnel; maintained full regulatory compliance and earned the highest performance grade (A*) plus special commendation from the supervising engineer

KEY PROJECTS & ACHIEVEMENTS

Sanofi 2025 Device Innovation Challenge - Global 4th Place Finalist

Dec. 2025

Team SanPump · 86 entries from 114+ universities globally · Paris, France

- **Led SolidWorks-driven CAD concept development** of a patient-centric wearable drug-delivery device: designed mechanical housing components, fluid path geometry, assembly interfaces, and external enclosure with attention to plastic component manufacturability, safety, and user ergonomics
- **Developed a data-driven mathematical model** applied to clinical measurement data to automatically adjust drug flow rate estimation in real time, demonstrating applied engineering analysis and calculation capability
- **Produced technical specifications and design documentation** for a multi-component wearable system; collaborated with teammates across mechanical design, clinical engineering, and digital modelling disciplines
- **Placed 4th globally from 86 competitive entries of 400+ Students**; team awarded \$50,000 - validated ability to deliver CAD-based mechanical design and prototype-ready engineering output under high-expectation deadlines in a multinational team

META 4.0 Summer School Design Challenge - 1st Place

Jul. 2024

Politecnico di Torino · Italy

- **Led a team of five** from initial brief through SolidWorks modelling, material selection, and physical prototype production of biodegradable 3D-printed manufacturing merchandise
- **Awarded Best Merchandise** - demonstrated rapid design-to-prototype execution, sustainability-informed material choices, and the ability to communicate design rationale clearly under competitive evaluation

AIIM - FEM-PINN Surrogate for Nonlinear Solid Mechanics

March 2026

Independent Research Project

- **Built an elastoplastic FEM solver in Python** modelling large-deformation structural behaviour; trained a Physics-Informed Neural Network (PINN) on 7,500 FEM outputs to predict full-field stress, strain, and displacement at 30–50x simulation speed-up
- **Developed an interactive engineering dashboard** to compare FEM and AI surrogate predictions - showcasing initiative in connecting numerical simulation to accessible digital engineering tools

International Product Safety Mentorship Programme - Pioneer Mentee

Oct. 2024

ICPHSO / European Commission · Brussels, Belgium

- **Selected as one of 11 mentees globally**; attended International Product Safety Week in Brussels, building knowledge of industrial product safety standards, regulatory compliance frameworks, and sustainable design practices across EMEA

PROFESSIONAL CERTIFICATIONS

SAP Certified Associate - S/4HANA Cloud PE, PP & Manufacturing (Dec. 2025)	Applied AI Lab: Deep Learning for Computer Vision - WorldQuant University (Oct. 2025)
Lean Six Sigma Black Belt - Alison / CSSC (Oct. 2025, 94%)	Lean Manufacturing: Principles & Implementation - Alison (Apr. 2025, 100%)
Total Productive Maintenance (TPM) - LinkedIn Learning (Oct. 2025)	Manufacturing Operational Excellence - Alison (Nov. 2024, 100%)
Project Management Essentials Certified - MSI (Oct. 2025)	Digital Manufacturing & Industry 4.0 - Alison (Mar. 2023, 100%)
IWCF Level 1 Well Control - International Well Control Forum (Aug. 2022)	Sustainability for Design, Construction & Manufacturing - LinkedIn Learning (Nov. 2025)

LANGUAGES

English C2 Fluent - IELTS Academic 7.0 (Reading 8.5)
French A2 Working Proficiency - CEFR certified, Ecole des Mines de Saint-Etienne
Isoko Native · **Yoruba** Basic working knowledge

PUBLICATIONS

7 peer-reviewed journal articles in computational manufacturing, machining simulation, advanced materials, and sustainable energy. Full publication list available on request.

AWARDS & SCHOLARSHIPS

- **Erasmus Mundus Joint Master’s Scholarship** - European Commission, top 1% globally
- **Sanofi 2025 Device Innovation Challenge** - 4th Place Global Finalist, \$50,000 team award (Paris, Dec. 2025)
- **Motor Valley Accelerator Hackathon Finalist** - Milan, Italy (Apr. 2024)
- Delta State Government Bursary & Monday Onyeme Foundation Scholar

RECOMMENDATIONS

Recommendation letters and references are available on request and on the Portfolio link above